

Neuromorphic Human-Computer Interaction: A Wikipedia Study

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ABSTRACT

I expect we write this section last. Lorem ipsum bla bla bla

CCS CONCEPTS

• **Computer systems organization;**

KEYWORDS

Neuromorphic Computing, Some more keywords

1 INTRODUCTION

This is the introduction section

2 BACKGROUND

In this section we provide an overview of background literature in current academic field.. Through this, we aim to embed our project in this field, and present the foundations of our research.

The idea that Wikipedia contains information on semantic relationships is not a new one. Several works investigate the possibility of extracting additional information on the semantic distance between concepts from Wikipedia, all of these show promising results[3–5]. These findings suggest a relationship between the semantic similarity of pages, and their distance in the Wikipedia graph-structure. [extend?]

The papers mentioned above mostly focus on information that can directly be extracted from the structure of data within Wikipedia. However, there are other aspects to consider as well, especially the way this Wikipedia data-structure is actually utilized by its users. West et al. [6] investigate the possibility of extracting semantic distances from a dataset of Wikispeedia games. This dataset consists of network traversal paths, where each path represents a person playing the Wikipedia game. In this game, a user is tasked to navigate from source-page A to goal-page B in the least amount of steps, thus these traces show the behaviour of targeted navigation on the Wikipedia network. This paper presents several interesting findings and insights, first and foremost that these traces produce semantic distances between concepts that are sensible when presented to human evaluators, improving over the LSA baseline. A second interesting observation they make is the fact that wayfinding for this task consists of two phases, a generalization phase and a specification phase. Since source-page A and goal-page B don't necessarily have a clear human-sensible path between the two directly, players of the game often first navigate to a "hub"-page.

Test

LLMs can do wikipedia traversal. Why? Semantic distance indicative of distance? [6] Paper Ana -> idea that there is a relationship between time/clicks and semantic distance [11][10]

Wikipedia traversal overview [7, 8] + "However, individuals' performance when navigating a knowledge space is influenced by personal characteristics, with younger and multilingual individuals demonstrating higher navigation abilities (Zhu, Yasseri, & Kertész, 2024)." + hypothesis to state that this suggests that internal bayesian might be closer to optimal path rather than semantic distance only -> younger people have more knowledge of the possible network (or think about it like this more)

Fits law -> expect somewhat of a similar relationship + log + the semantic/optimal distance as distance, width analogous to ease of access + Intuition because of the hub behaviour

LLM to find semantic distances: - why? - what llm?

Network traversal to determine semantic distance:

Semantic distance as a predictor of click behaviour: + tf-idf rather than title similarity - but not insane to use titles [9] + interesting note on structure

Might be interesting:

Specifically, good to see people combine information from wikipedia into semantic distance already. New insights might be learned from user behaviour, and inversely we might learn more about human behaviour:

3 METHODS

In order to try and find useful difficulty metrics for Wikispeedia, we compared various potential metrics to see how well they predict the 'real' difficulty, measured by how long the user ended up taking. To do this, we tried the same formula as Fitts' law, $ID = \log_2(\frac{D}{W} + 1)$, by investigating metrics that are conceptually similar to distance (how far we are removed from our target page) and width (how easy it is to reach the target page).

Furthermore, we binned together tasks with similar difficulty. The reason for this is that without binning, we have a lot of tasks

3.1 Distance metrics

In this paper, we tried the following three distance metrics: *semantic distance*, *semantic graph distance*, and *graph distance*.

Semantic distance. The first distance metric, *semantic distance*, is calculated by the LLM transformer model [?] and indicates the similarity of the titles of two Wikipedia articles.

Semantic graph distance. In order to calculate the second distance metric, *semantic graph distance*, we first construct a graph of all Wikipedia pages in the dataset. The articles represent the vertices in the graph, and the links between them are the edges. We then use the LLM's semantic distance as the length of each edge. The semantic graph distance between two articles is the length of the shortest path in this weighted graph.

Graph distance. The last distance metric, *graph distance*, represents the length of the shortest path between two pages, but where all edges simply have a weight of 1, representing the minimal number of clicks.

3.2 Width metrics

Additionally, we tried two width-analogous metrics: *PageRank* and *indegree*.

PageRank. To explain the intuition behind PageRank, assume a user who is surfing Wikipedia. This user starts on some random page and randomly clicks a link on the wikipedia page. At some point this user gets bored and starts at some other random page and continues clicking links randomly. The PageRank of page a is defined as the chance this user reaches page a [2]. Therefore, the higher the PageRank, the easier it is to reach the article, similar to how a wider button is easier to reach.

Indegree. The second width metric is *indegree*. This is the number of links to the target article in the Wikipedia graph. The idea behind this is that, if there are more incoming links, we can more easily reach the target page.

4 RESULTS

We present the results in two figures, each containing several graphs. Figure 1 gives graphs that plots the *ID* against the number of clicks of a game and Figure 2 gives graphs that plot the *ID* against the duration of a game.

The first row uses semantic distance, the second row semantic graph distance, the third row uses the graph distance, and the fourth row uses the value 1 for the distance. The first column uses the value 1 for the width, the second column PageRank and the third column the indegree. The value 1 for the width and the distance serve as a control metric to see how they influence the correlation. The bottom left graph is empty, since they have the value 1 for both distance and correlation, which always gives the same *ID*.

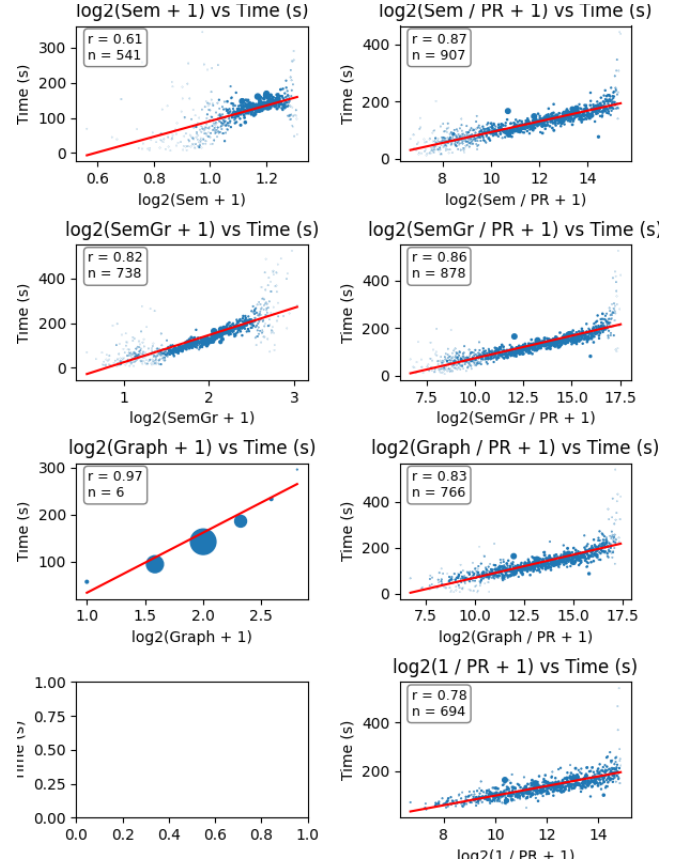


Figure 1: caption

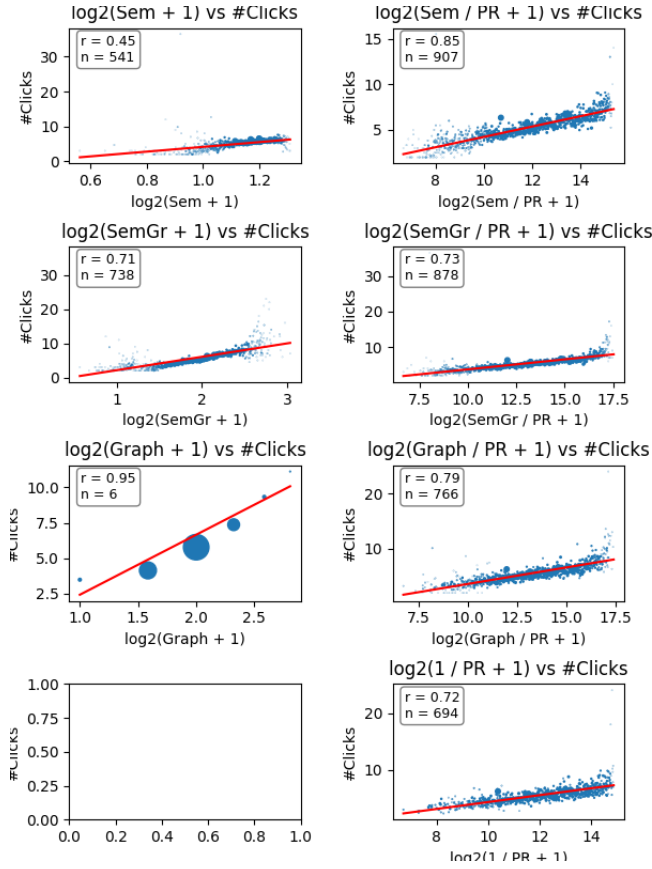


Figure 2: caption

Some examples. A paginated journal article [1], an

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A ADDITIONAL INFORMATION

This is the appendix